Supplementary Material for "Realistic Loss Budgeting and Defect Tolerance in Lead-Free $RbGeI_3$ Thin-Film Solar Cells"

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September 30, 2026

S1. One-at-a-time sweep details

Full per-parameter traces for the eleven sequential optimization steps summarized in Section 2 of the main text (baseline campaign; consolidated device 26.81%).

Absorber thickness (Fig. S1)

PCE peaks at 0.7 μm (20.70%); J_{SC} climbs 28.14→34.11 mA/cm^2 while V_{OC} slides 0.823→0.767 V. The upper J_{SC} readings come from the early Step-1 deck with extended sub-gap absorption; re-simulation on the consolidated uniform-1.4 eV deck gives 25.2–31.3 mA/cm^2 with the same peak-plateau shape, so 0.7 μm stands.

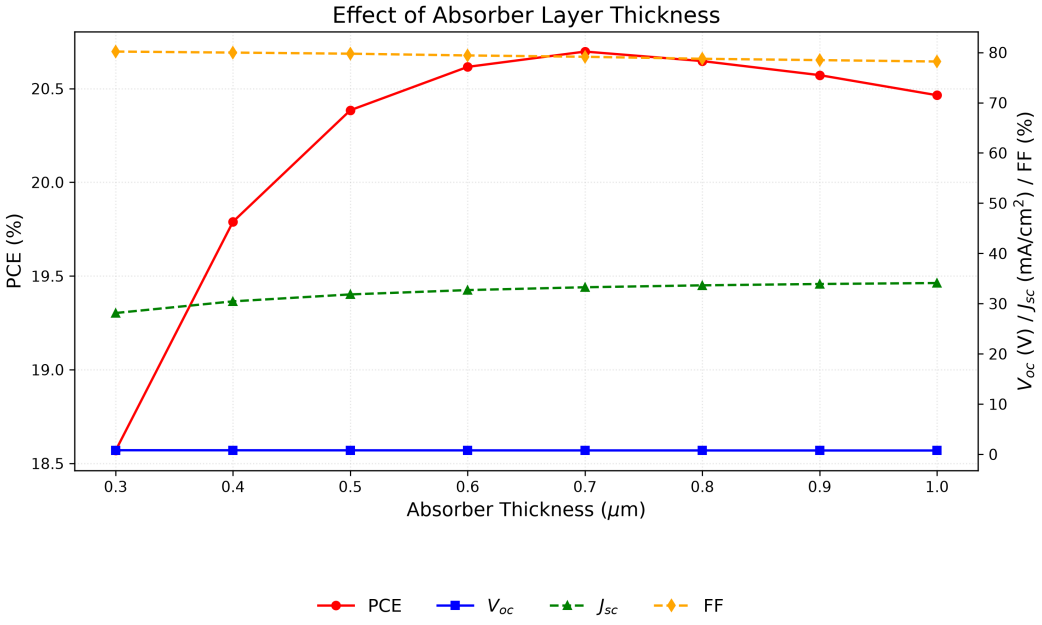


Figure S1: Fig. S1: PCE, V_{OC}, J_{SC}, FF vs absorber thickness.

Dielectric constant (Fig. S2)

Raising ϵ_r 15→23.1 moves PCE 19.92→19.79% via V_{OC} only; $\epsilon_r = 15$ kept.

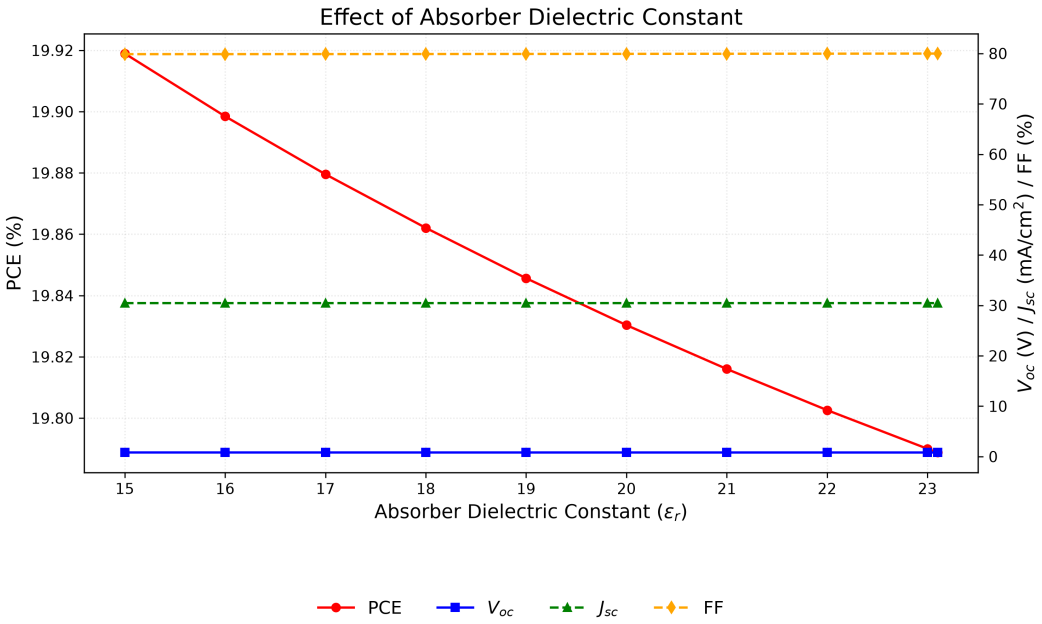


Figure S2: Fig. S2: sweep over absorber dielectric constant.

Electron affinity (Fig. S3)

Peak at $\chi = 3.9\,\text{eV}$ (19.79%); 4.2 eV costs V_{OC} 0.812→0.710 V at flat J_{SC}.

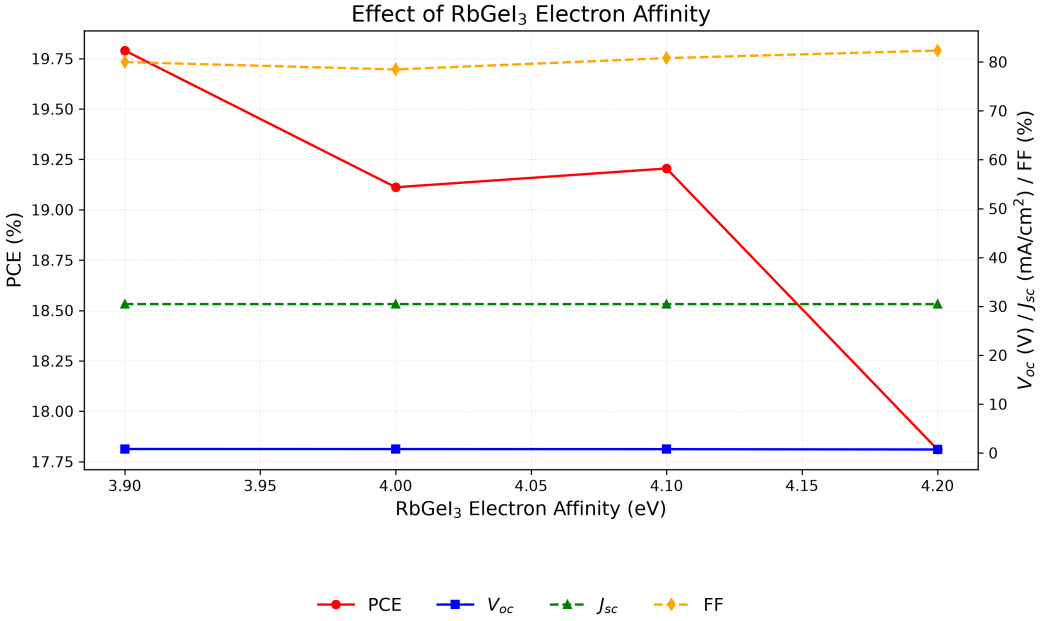


Figure S3: Fig. S3: sweep over absorber electron affinity.

TiO_2/$RbGeI_3$ interface (Fig. S4)

The performance cliff: $10^{12}\rightarrow10^{18}\,\mathrm{cm}^{-2}$ sends PCE 21.04→8.40%; $10^{14}\,\mathrm{cm}^{-2}$ adopted as the fabricable target.

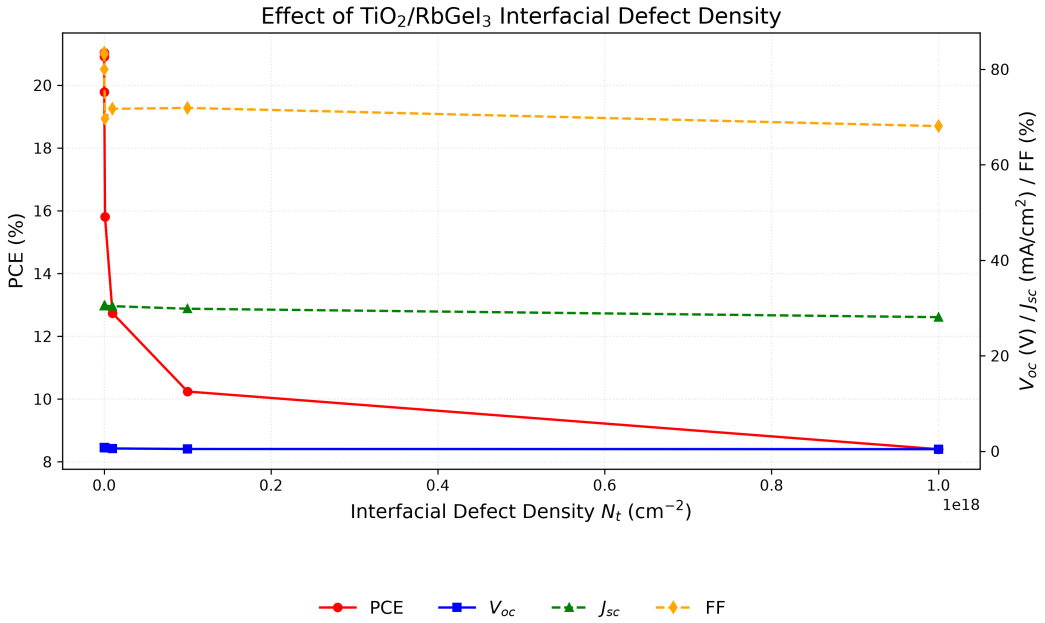


Figure S4: Fig. S4: TiO_2/$RbGeI_3$ interfacial defect sweep.

$RbGeI_3$/CuI interface (Fig. S5)

Far more tolerant (19.79→18.19%); the electron side pins the limit. $10^{14}\,\mathrm{cm}^{-2}$ adopted for consistency.

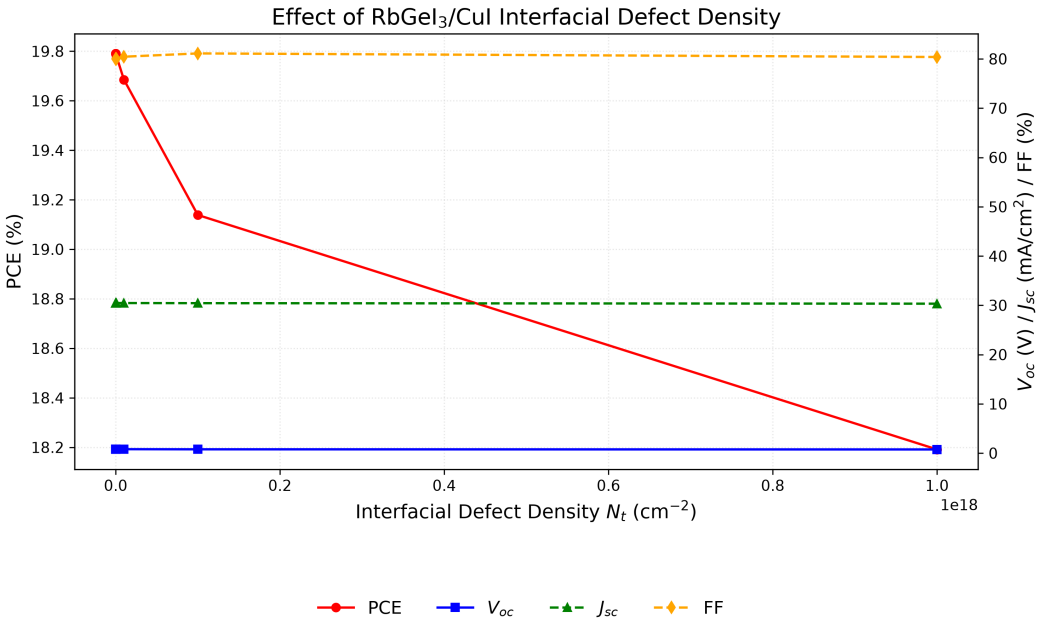


Figure S5: Fig. S5: $RbGeI_3$/CuI interfacial defect sweep.

Bandgap (Fig. S6)

V_{OC} 0.840→1.002 V vs J_{SC} 33.59→20.30 mA/cm^2 over 1.3–1.7 eV; optimum 1.4 eV sits 0.06 eV from the SQ optimum (≈1.34 eV); $\lambda_c = 1240/1.4 \approx 886$ nm matches the QE edge.

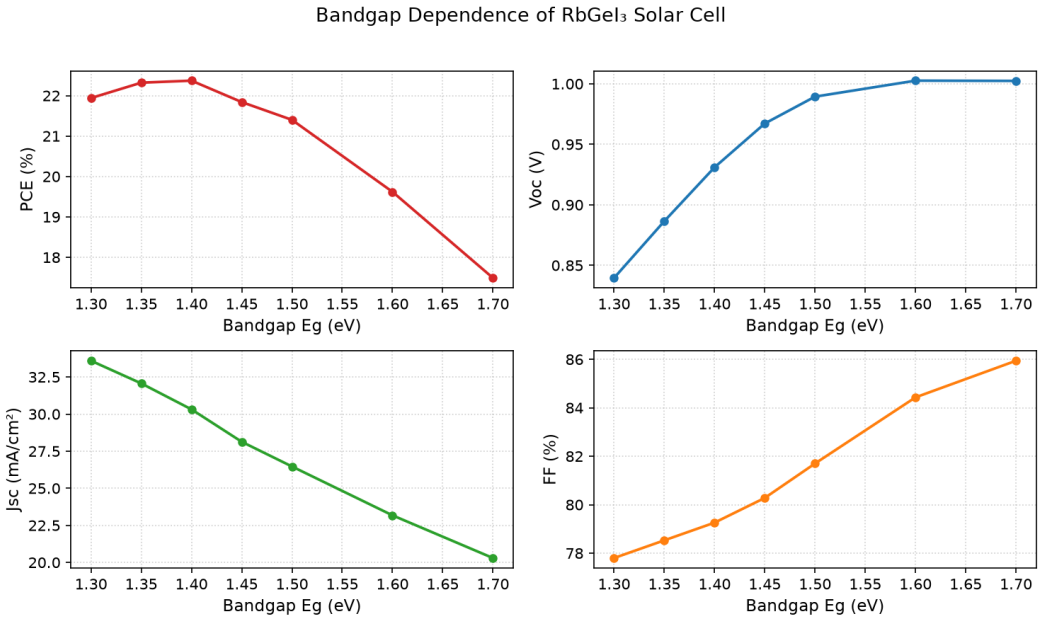


Figure S6: Fig. S6: sweep over absorber bandgap.

TiO_2 thickness (Fig. S7)

Best at the thinnest ETL (21.12% at 10 nm); thicker layers add resistance and parasitic absorption.

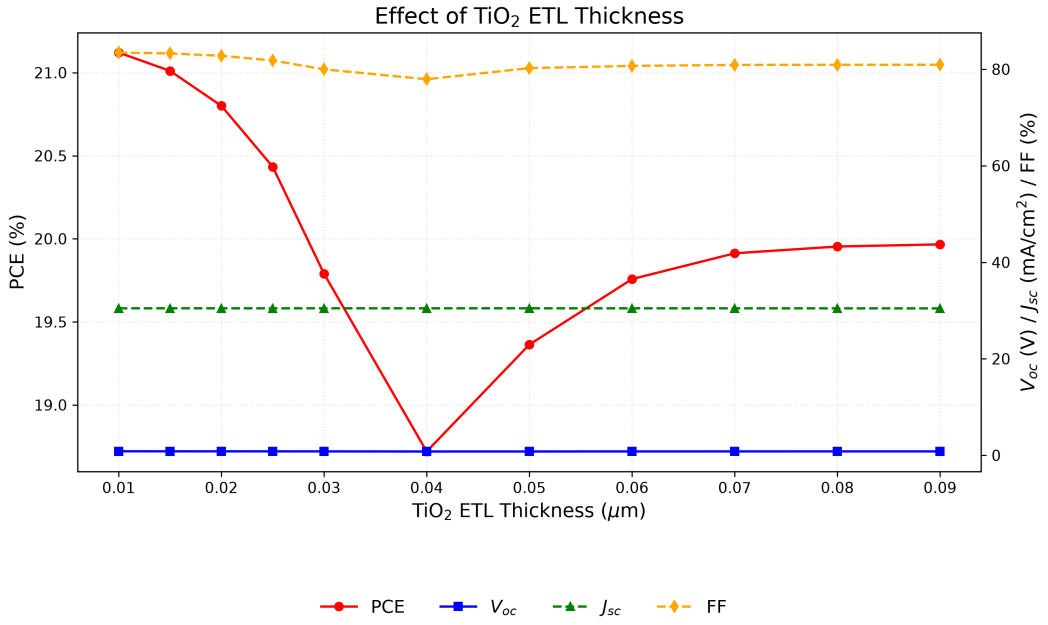


Figure S7: Fig. S7: sweep over TiO_2 ETL thickness.

CuI thickness (Fig. S8)

Flat within 0.002 pp over 0.1–1.9 μm; 100 nm minimizes material use.

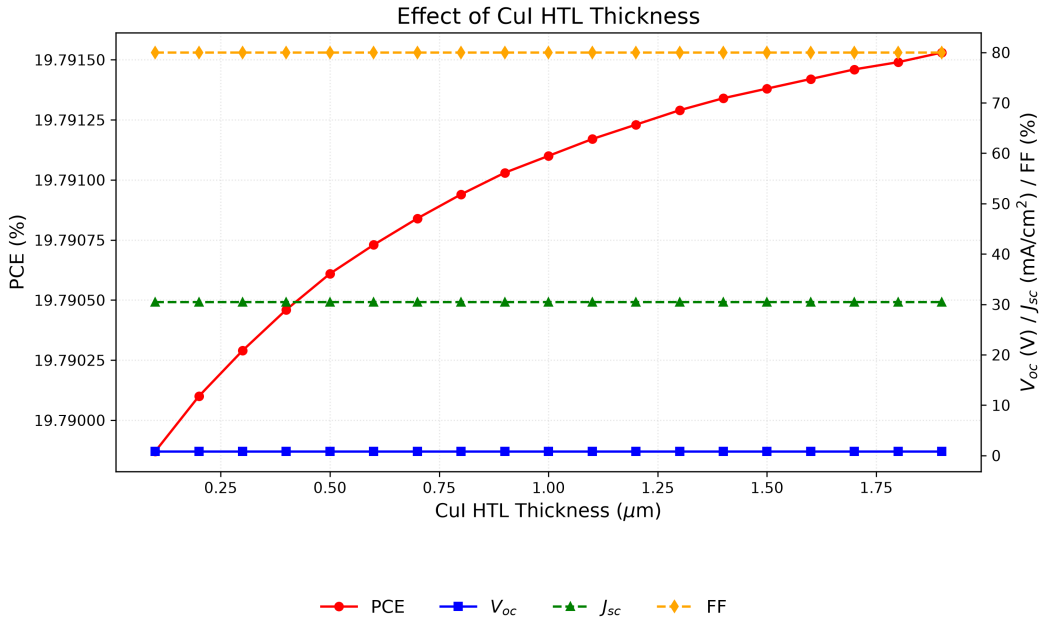


Figure S8: Fig. S8: sweep over CuI HTL thickness.

N_C and N_V (Figs. S9–S10)

Lower densities raise V_{OC} at flat J_{SC}; both carried at $10^{17}\,\mathrm{cm}^{-3}$ (PCE 20.65% and 23.54% at the sweep bottoms).

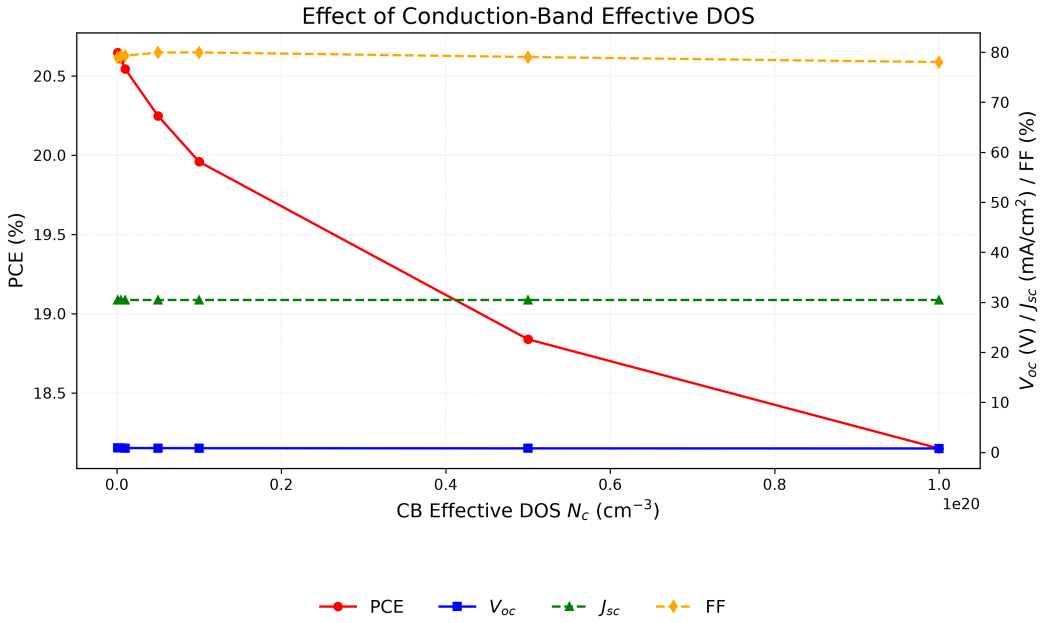


Figure S9: Fig. S9: sweep over N_C.

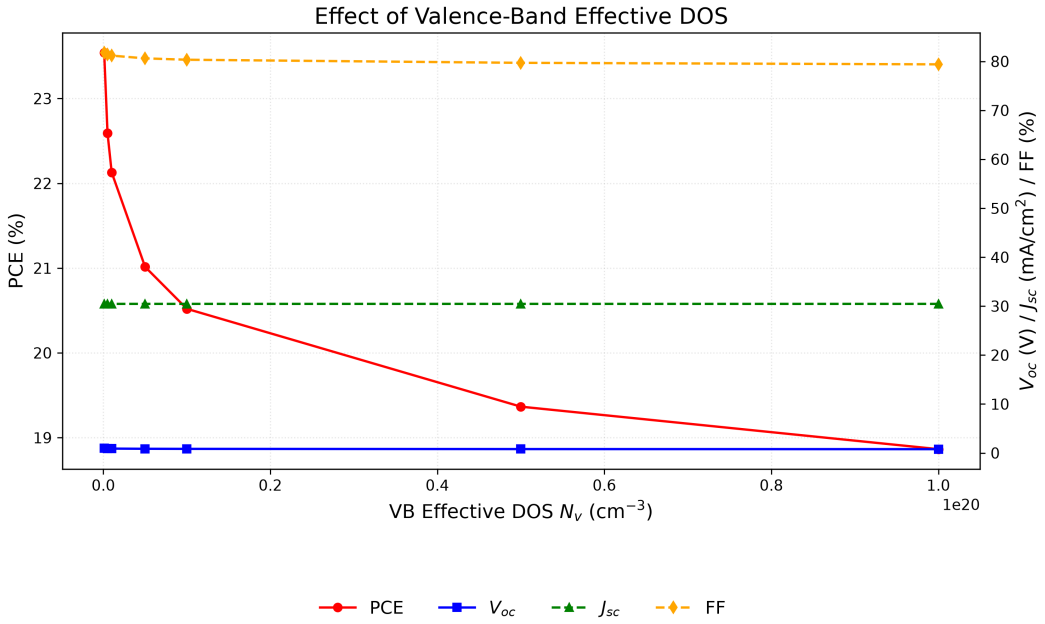


Figure S10: Fig. S10: sweep over N_V.

S2. Illumination and dark characteristics

0.1–1.5 suns: J_{SC} linear in intensity, V_{OC} logarithmic with ideality $n \approx 1.2$ (Fig. S11). Dark J–V rectifies 1.8×10^{10} with leakage below 10^{-10} A/cm^2 (Fig. S12).

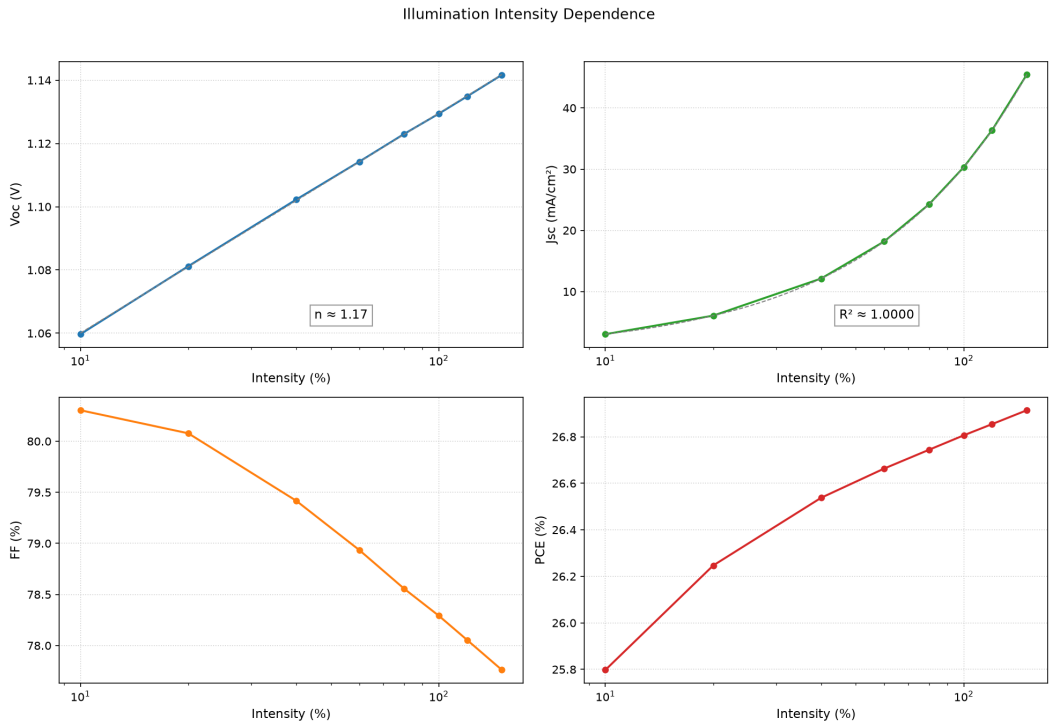


Figure S11: Fig. S11: illumination dependence.

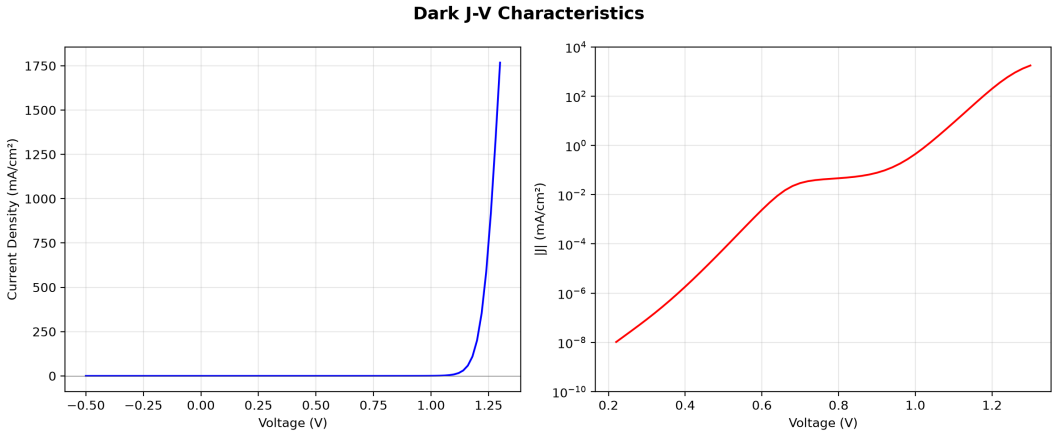


Figure S12: Fig. S12: dark J–V.

S3. Tables

Study	Device	PCE (%)	Key limitation
Pindolia 2022	FTO/TiO_2/$RbGeI_3$/NiO/Ag	9.77	Modest baseline; oxidation unaddressed
Saikia 2022	FTO/TiO_2/$CsGeI_3$/CuI/Au	10.80	No multi-HTL comparison
Loumachi 2024	ITO/C_{60}/$RbGeI_3$/CBTS/Ag	27.21	Interface defects unaddressed
Raj 2025	FTO/TiO_2/$RbGeI_3$/PEDOT:PSS/Au	18.60	Organic HTL stability
Pathak 2026	FTO/TiO_2/$RbGeI_3$/Spiro/Au	26.47	Single ETL/HTL; oxidation unaddressed
Dash 2025	DFT+SCAPS $RbGeX_3$	25.76	Material-level focus

Table S1: Table S1: SCAPS literature context for Ge-based cells.

Parameter	TiO_2	FTO	$RbGeI_3$	CuI
Thickness (nm)	10	500	700	100
E_g (eV)	3.2	3.2	1.4	3.1
χ (eV)	4.0	4.4	3.9	2.1
ϵ_r	9	9	15	6.5
N_C (cm^{-3})	2×10^{18}	2.2×10^{18}	1×10^{17}	2.8×10^{19}
N_V (cm^{-3})	1.8×10^{19}	1.8×10^{19}	1×10^{17}	1×10^{19}
μ_e (cm^2/Vs)	20	20	28.6	100
μ_p (cm^2/Vs)	10	10	27.3	43.9
N_D (cm^{-3})	9×10^{16}	1×10^{19}	1×10^{9}	0
N_A (cm^{-3})	0	0	1×10^{9}	1×10^{18}
N_t (cm^{-3})	1×10^{15}	1×10^{14}	1×10^{14}	1×10^{14}

Table S2: Table S2: consolidated (optimized) layer parameters. Interfaces: $10^{14}\,cm^{-2}$ both junctions (Table S3 in prior submission).

BSF d (nm) × N_A (cm^{-3})	10^{17}	10^{18}	10^{19}
30	28.61	30.07	30.35
50	28.74	30.07	30.35
80	28.76	30.08	30.35
100	28.76	30.08	30.35

Table S3: Table S3: BSF grid PCE (%; `r6_bsf_sweep.json`, 12/12 converged). Doping dominates; thickness flat.

Waterfall node	V_{OC} (V)	J_{SC}	$PCE_{1\text{-sun}}$ (%)
R0 ideal	1.1298	30.32	26.81
R1 +radiative/Auger	1.1294	30.32	26.79
R2 +10% reflection	1.1263	27.28	24.08
R3 +bulk 10^{15}	1.1083	27.28	22.34
R4 +interfaces 10^{15}	1.0473	27.26	20.80
R5full BSF + full realism	1.0877	27.27	23.29

Table S4: Table S4: native loss waterfall (`r6_realistic_waterfall.json`).

References

[1] W. Shockley and H. J. Queisser, J. Appl. Phys. 32, 510–519 (1961).

[2] S. Rühle, Sol. Energy 130, 139–147 (2016).

[3] C. H. Henry, J. Appl. Phys. 51, 4494–4500 (1980).